MD ASHFAK ALAM

Generative AI Engineer

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PROFESSIONAL SUMMARY

Generative AI Engineer with **2 years** of experience building and deploying **LLM applications**, **RAG pipelines**, and **multi-agent systems** using **Python**, **LangChain**, **LangGraph**, and **FastAPI**. Built RAG over **1,000+ documents** with **FAISS**, **ChromaDB**, and **Pinecone**, reaching **90%** answer relevance and reducing hallucinations and latency through retrieval tuning. Developed **intelligent agents** with **tool calling** that automated **8+ business processes** and significantly cut turnaround time. Skilled in **prompt engineering**, **LLM evaluation**, and benchmarking models (**GPT-3.5/4**, **Llama 3**) on quality, latency, and cost to select the best fit for each use case. Experienced in production delivery: **FastAPI** services, **Docker** on **AWS**, **CI/CD**, and **LangSmith** monitoring, sustaining **99%** uptime across environments. Owned **production support**, **API testing**, and **error handling**, working with client and engineering teams to deliver reliable **Generative AI** solutions.

TECHNICAL SKILLS

Generative AI & LLMs	Large Language Models (LLMs), Generative AI, GPT-3.5/4, Llama 3/3.2, Prompt Engineering, RAG (Retrieval-Augmented Generation), AI Agents, Agentic AI, Multi-Agent Systems, Tool Calling, Function Calling, Structured Outputs, Conversation Memory, Embeddings, Semantic Search
LLM & AI Frameworks	LangChain, LangGraph, LangSmith, Hugging Face Transformers, OpenAI API, Groq API, Ollama
RAG & Vector Databases	FAISS, ChromaDB, Pinecone, Vector Search, Semantic Retrieval, Metadata Filtering, Document Processing
LLM Evaluation & Optimization	LLM Evaluation, LLM Benchmarking, RAG Evaluation, Prompt Optimization, Hallucination Reduction, Latency and Cost Optimization
Production & MLOps	Production Deployment, MLOps Pipelines, CI/CD, Model Monitoring (LangSmith), API Testing, Error Handling, Debugging, Performance Validation, Production Support
Backend & APIs	FastAPI, REST APIs, API Integration, Streamlit, Postman
Cloud & DevOps	Docker, AWS (EC2, S3), Git, GitHub
Languages & Databases	Python, SQL, Java, MySQL, PostgreSQL, NumPy, Pandas, Requests, BeautifulSoup, OOP, Data Structures & Algorithms
Other	React.js, Redux Toolkit, Tailwind CSS, JavaScript (ES6+)

PROFESSIONAL EXPERIENCE

Cynob IT Consultancy, Hyderabad | *Generative AI Engineer* **October 2024 – Present**

- Built and deployed production **RAG pipelines** (**embeddings**, **FAISS**/**ChromaDB**/**Pinecone**, **semantic search**) over **1,000+** documents; tuned chunking, top-k, and **metadata filtering** to reach **90%** answer relevance, cutting latency by **35%** and reducing hallucinated answers.
- Developed **AI agents** with **LangChain** and **LangGraph** using **tool calling** and multi-step workflows, automating **8+** business processes and cutting turnaround time by **45%**.
- Improved **LLM** response accuracy through **prompt engineering**, structured outputs, and context management; benchmarked **GPT-3.5/4** and **Llama 3** on quality, latency, and cost to pick the best model per use case.
- Ran **LLM evaluation** across multiple models and monitored LLM applications with **LangSmith** in **MLOps pipelines**, tracking accuracy, latency, and cost to drive a **20%** cut in inference cost.
- Deployed RAG and agent services as production **REST APIs** with **FastAPI**, serving **100+** users with low response times; ran **API testing**, **error handling**, and performance validation before each release.
- Containerized LLM services with **Docker** and deployed on **AWS** (EC2, S3) using **CI/CD**, shortening deployment time and sustaining **99%** uptime.
- Owned **production support**: troubleshot runtime errors, integration failures, and application issues through **debugging** and root-cause fixes, cutting production errors by **40%**.
- Collaborated with data engineers, ML engineers, and client stakeholders to translate business problems into technical requirements, delivering Generative AI solutions and supporting pre-sales demos.

PROJECTS

Intellica | *Autonomous Multi-Agent Research System* GitHub

Python, LangChain, Hugging Face (GPT-OSS-120B), Tavily API, Requests, BeautifulSoup, Streamlit

- Built a **multi-agent research system** with **Reader**, **Writer**, and **Critic** agents using **LangChain** and **GPT-OSS-120B** that turns user queries into structured, source-backed research reports.
- Implemented a **web research pipeline** with **Tavily API**, **Requests**, and **BeautifulSoup** to search the web, select the **top 3** sources, and extract relevant content as grounded context for report generation.
- Designed an end-to-end **agentic workflow** (search tool, scraping, Writer Agent, Critic Agent) that passes search results and scraped content as grounded context, so reports stay tied to cited sources.
- Added **Critic Agent** checks for source quality, factual consistency, relevance, and unsupported claims (**hallucination check**), improving research reliability by **30%**.
- Developed a ChatGPT-style **Streamlit** interface with chat history, source URLs, raw research results, critic feedback, dark/light mode, and **Markdown report export**, cutting manual research effort by **50%**; managed API keys securely with **python-dotenv**.

Summify AI | *Meeting and Video Summarizer* GitHub

Python, LangChain, Ollama (Llama 3.2), Whisper, Sarvam AI, RAG, Streamlit

- Built an AI app using **Llama 3.2** (Ollama) and **LangChain** that converts videos into titles, summaries, action items with owners and deadlines, key decisions, and open questions, cutting manual note-taking time by **70%**.
- Created an audio pipeline (**yt-dlp**, **Pydub**, 16kHz mono chunking) with **Whisper** (English) and **Sarvam AI** (Hinglish), supporting **YouTube videos** and local media files across **2 languages**.
- Implemented a **RAG** pipeline (vector store, retriever, LLM) so users can chat with their meeting and get answers grounded only in the transcript, with **90%+** answer accuracy.
- Applied **prompt engineering** and structured output generation for consistent insights across **50+** video and audio inputs.
- Delivered a **Streamlit** interface with English and Hinglish language selection, dark/light mode, and transcript-based Q&A; managed configuration and API keys with **python-dotenv**.

EaseKart | *eCommerce Frontend Application* GitHub

React.js, Redux Toolkit, Context API, Tailwind CSS, JavaScript ES6+, REST APIs, Figma, Git

- Developed a responsive **React.js** eCommerce frontend with product listings, cart management, and reusable UI components, using **Redux Toolkit** for state, **Context API** for theming, and **Tailwind CSS** across **3+ device sizes**.
- Integrated **REST APIs** for dynamic product data and translated **Figma designs** into accessible, mobile-first interfaces.

EDUCATION

Master of Computer Applications (MCA) | A.P.J. Abdul Kalam University, Indore **June 2022 – May 2024**

Bachelor of Science in Physics (Hons.) | Bhimrao Ambedkar Bihar University, Bihar **June 2016 – May 2019**

CERTIFICATIONS & COURSES

Complete Generative AI Course with LangChain and LangGraph | Krish Naik, KRISHAI Technologies

Python Programming | CodeWithHarry